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Structural Variation in the Human Genome and its Role in Disease

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genomic disorders, array CGH, copy-number variants, molecular mechanisms of genomic rearrangements, microdeletions and microduplications

Abstract

During the last quarter of the twentieth century, our knowledge about human genetic variation was limited mainly to the heterochromatin polymorphisms, large enough to be visible in the light microscope, and the single nucleotide polymorphisms (SNPs) identified by traditional PCR-based DNA sequencing. In the past five years, the rapid development and expanded use of microarray technologies, including oligonucleotide array comparative genomic hybridization and SNP genotyping arrays, as well as next-generation sequencing with "paired-end" methods, has enabled a whole-genome analysis with essentially unlimited resolution. The discovery of submicroscopic copy-number variations (CNVs) present in our genomes has changed dramatically our perspective on DNA structural variation and disease. It is now thought that CNVs encompass more total nucleotides and arise more frequently than SNPs. CNVs, to a larger extent than SNPs, have been shown to be responsible for human evolution, genetic diversity between individuals, and a rapidly increasing number of traits or susceptibility to traits; such conditions have been referred to as genomic disorders. In addition to well-known sporadic chromosomal microdeletion syndromes and Mendelian diseases, many common complex traits including autism and schizophrenia can result from CNVs. Both recombination- and replication-based mechanisms for CNV formation have been described.

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INTRODUCTION

In the past 15 years, it has become evident that higher-order genomic architectural features can lead to a susceptibility to DNA rearrangements that are frequent causes of disease. Such conditions have been referred to as genomic disorders (1, 2). Genomic rearrangements can cause Mendelian diseases or complex traits such as behaviors, or they can represent benign polymorphic changes. A major mechanism by which rearrangements convey phenotypes is gene dosage (3) due to a variation in either gene copy number or essential regulatory elements; when the deleted or duplicated region harbors a dosage-sensitive genomic interval, the rearrangement can lead to an abnormal phenotype.

Genomic disorders are often sporadic, caused by de novo rearrangements. Locus-specific mutation rates for genomic rearrangements range between 10^{-4} and 10^{-5}, at least 1000- to 10,000-fold more frequent than point mutations (4). In general, genomic disorders occur with a similar frequency; however, significant differences in incidences in different world populations have been observed. In some such cases, population-specific structural variations of the genomic region in the patients' parents have been found, demonstrating that the variation of genomic architecture can be a significant factor in disease susceptibility.

Copy-Number Variation

The recent expanded application of genome-wide analysis tools such as comparative genomic hybridization (CGH) to analyze human genomes has led to the discovery of extensive genomic structural variation, ranging in size from kilobases (kb) to megabases (Mb), that are not identifiable by conventional chromosomal banding. These changes have been termed copy-number variations (CNVs) (5–7).

Deletions, duplications, triplications, insertions, and translocations can all result in CNVs. In addition, balanced genomic inversions leading to DNA structural variations that do not cause CNV can nevertheless contribute significantly to genome instability. Despite extensive studies, the total number, position, size, gene content, and population distribution of CNVs remain elusive. CNVs of 1–50 kb in size have been under-ascertained because there has not been an accurate molecular method available to study these smaller rearrangements on a genome-wide scale in different populations (7, 8). Recent analyses revealed 11,700 CNVs overlapping over 1000 genes (7, 9). CNVs may account for 13% of the human genome. It is anticipated that the wider application of array CGH techniques and next-generation sequencing (with e.g. paired-end methods and "read number" as surrogate measures for copy number) will reveal a significantly larger scale of structural variation among different individuals and populations, as the majority of CNVs appear smaller in scale and are beyond the resolving capability of current arrays (10, 11).

CNVs can be inherited or sporadic; large de novo CNVs are thought more likely to be disease causative. However, the phenotypic effects of CNVs are sometimes unclear and depend mainly on whether dosage-sensitive genes or regulatory sequences are affected by the genomic rearrangement. Sporadic disease can also result from a combination of two CNVs at a single locus (e.g., autosomal recessive spinal muscular atrophy and nephronopthisis) or theoretically from two or more CNVs at different loci from two normal parents (12).

CNVs have been cataloged in public databases such as the Toronto Database of Genomic Variants. Clinically relevant CNVs can be found in DECIPHER (DatabasE of Chromosomal Imbalance and Phenotype in Humans using Ensembl Resources, **https://decipher.sanger.ac.uk**) (13) and ECARUCA (European Cytogeneticists Association Register of Unbalanced Chromosome Aberrations, **http://agserver01.azn.nl:8080/ecaruca/ecaruca.jsp**).

Low-Copy Repeats

Up to 5% of the reference haploid human genome is present in two or more copies (14).

Structural variation: a change of genomic DNA greater than 1 kb in size that distinguishes two genomes in one species. Structural variations can be either unbalanced (CNVs) or balanced (e.g., genomic inversions, reciprocal translocations). Both types of structural variation require a break in the DNA phosphodiester backbone

CGH: comparative genomic hybridization

Copy-number variation (CNV): a segment of DNA ranging from one kilobase to several megabases in size that represents an imbalance between two genomes from one species. CNVs perturb the normal biological balance of the diploid state at any given locus. Deletions, duplications, triplications, insertions, or translocations can result in CNVs

DNA fragments >1 kb in size and of >90% DNA sequence identity have been termed low-copy repeats (LCRs) or segmental duplications. Many LCRs in the human genome have complex structure, and many have arisen during primate speciation over the past 25–40 million years as a result of serial segmental duplications. LCRs longer than 10 kb and of over ~97% sequence identity can lead to local genomic instability. LCRs have been shown to stimulate and/or mediate constitutional (i.e., inherited; both recurrent and nonrecurrent), evolutionary, and somatic genomic rearrangements (15, 16). LCRs can cause genomic instability and either mediate or stimulate CNV formation. Furthermore, LCRs can themselves vary in copy number.

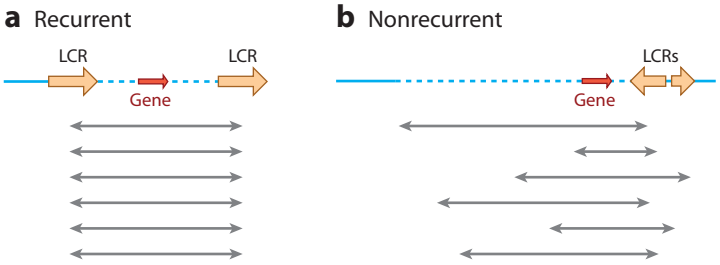


Figure 1

Schematic representation of the recurrent and nonrecurrent genomic rearrangements associated with genomic disorders. (*a*) The common recurrent same-sized deletions and duplications have both breakpoints mapping (clustering) within the directly oriented low-copy repeats (LCRs). The vast majority of the recurrent rearrangements result from nonallelic homologous recombination (NAHR). (*b*) The nonrecurrent rearrangements have breakpoints scattered. If a LCR is present, the breakpoints group in its vicinity. Most of these rearrangements result from FoSTeS/MMBIR (fork stalling and template switching/microhomology-mediated break-induced replication) mechanisms of formation. See text for details.

MECHANISMS FOR FORMATION OF COPY-NUMBER VARIATIONS

Nonallelic Homologous Recombination

When LCRs have DNA sequence identity greater than ~97% and are located at a distance less than ~10 Mb from each other, they can lead to misalignment of chromosomes or chromatids and mediate nonallelic homologous recombination (NAHR) that can result in unequal crossing-over (17), with recombination hotspots, gene conversion, and apparent minimal efficient processing segments. NAHR between directly oriented LCRs results in deletions or reciprocal duplications of the genomic segment between them, whereas NAHR between the inverted LCRs leads to an inversion of the intervening genomic segment (**Figure 1**). When LCRs have a complex structure consisting of both direct and inverted subunits, they can serve as NAHR substrates leading to, respectively, genomic deletions/duplications and inversions. This molecular mechanism has been shown to be responsible for the vast majority of the common-sized recurrent rearrangements—reciprocal deletions and duplications, or inversions.

Nonhomologous End Joining

Until recently, little has been known about nonrecurrent genomic rearrangements. They have been thought to arise by another recombination-based mechanism, nonhomologous end joining (NHEJ), responsible for repair of DNA double strand breaks. In NHEJ, double strand breaks are detected; then both broken DNA ends are bridged, modified, and finally ligated (18). In contrast to NAHR, NHEJ does not require LCRs or minimal efficient processing segments to mediate the recombination but may also be stimulated by genome architecture. The product of repair often contains additional nucleotides at the DNA end junction, leaving a "molecular scar" (15).

Replication-Error Mechanisms

In addition to NAHR and NHEJ, fork stalling and template switching (FoSTeS), a mechanism based on DNA replication error, has recently been shown to play an important role in the origin of genomic-disorder-associated nonrecurrent rearrangements that have a complex structure, e.g., deletions and/or duplications interrupted by either normal copy number or

LCR: low-copy repeat

Nonallelic homologous recombination (NAHR): the predominant molecular mechanism of formation of recurrent genomic rearrangements. The flanking LCRs are used as recombination substrates and both stimulate and mediate the NAHR event

NHEJ: nonhomologous end joining

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FoSTeS/MMBIR (fork stalling and template switching/ microhomology-mediated break-induced replication): replication-based mechanisms of DNA repair that utilize nucleotide microhomology at the rearrangement breakpoints and can account for the formation of complex genomic rearrangements

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triplicated genomic segments (19, 20). In this model, the DNA replication fork stalls; the lagging strand disengages from the original template and anneals to another replication fork in physical proximity, by virtue of microhomology at the 3′ end, "priming" or reinitiating DNA synthesis. FoSTeS resembles the long-distance template-switching model described in *E. coli* to explain the genesis of amplifications induced under stress (21).

A general model proposed to be responsible for nonrecurrent CNVs and complex genomic rearrangements in all life forms, based on experimental observations from human and other model organisms, is termed microhomology-mediated break-induced replication (MMBIR) (22). Depending on the location of the new fork (upstream or downstream), a deletion or duplication, respectively, will occur. Whether the lagging or leading strand in the new fork is used as a template determines if the erroneously incorporated fragment from the new replication fork will be in direct or inverted orientation with respect to its original position (22).

In addition to the Pelizaeus-Merzbacher disease region on chromosome Xq22 (19), the FoSTeS mechanism has been demonstrated subsequently in other genomic regions, e.g., *MECP2* duplication in Xq28 (23), deletions and duplications in 17p13.3 (24, 25), deletions and duplications in 17p11.2p12 (20), deletions and duplications in 9q34 (26), and variations in at least 17 other reanalyzed genes (20). It has been proposed that FoSTeS/MMBIR may be a major mechanism for generating structural variation, particularly nonrecurrent CNVs and complex genomic rearrangements, in the human genome and even for the formation of LCRs themselves (27, 28).

HOW COPY-NUMBER VARIATIONS CONVEY PHENOTYPES

Deletion and duplication rearrangements can cause a phenotype by several molecular mechanisms. One of the most common recognized mechanisms is altering the copy number of a gene (or genes) sensitive to a dosage effect (3).

CNVs can also result in a position effect (29). Such position effects have been documented for apparently balanced translocations and even exert these effects when the breakpoints map as far as ∼1 Mb away either upstream or downstream from the causative gene. Using array CGH, Lee at al. (30) reported a microduplication located downstream of the *PLP1* gene in a patient with spastic paraplegia type 2. Smyk et al. (31) described a 21-year-old 46,XY female manifesting primary amenorrhea, a small immature uterus, gonadal dysgenesis, and notably absent clinical findings consistent with adrenal insufficiency with a submicroscopic 257,782-bp deletion 11,320 bp upstream of *NR0B1*. The authors hypothesized that loss of regulatory sequences may have resulted in position-effect upregulation of *NR0B1* (*DAX1*) expression, consistent with phenotypic consequences of *NR0B1* duplication. A clinical array CGH revealed a gene-free 120,772-bp tandem duplication located 7326 bp 5′ to *NR0B1* (*DAX1*) in a patient with premature puberty and an enlarged ovary (P. Stankiewicz & S.-W. Cheung, personal communication). Dathe et al. (32) detected an ∼5.5-kb duplication in a noncoding sequence located ∼110 kb downstream of the *BMP2* gene in chromosome 20p12.2 in two families with brachydactyly type A2. The duplicated region contains highly evolutionarily conserved sequences suggestive of a long-range regulator. Klopocki et al. (33) identified a 588,819-bp duplication encompassing the previously described ZRS regulatory element located ∼1 Mb 5′ from the intact sonic hedgehog gene *SHH*, in intron 5 of the *LMBR1* gene in chromosome 7q36.3, in a large family with triphalangeal thumb and polysyndactyly. Moreover, Kurth et al. (34) reported an ∼2-Mb duplication of regulatory sequences in a gene desert upstream of *SOX9* in four unrelated families with brachydactyly-anonychia phenotype compatibile with Cooks syndrome (MIM 106995), characterized by onychodystrophy, anonychia, brachydactyly of the fifth finger, and digitalization of the

thumbs, with absence or hypoplasia of the distal phalanges of the hands and feet.

Another molecular mechanism by which rearrangements of the genome may convey or alter a disease phenotype is the unmasking of recessive mutations or functional polymorphisms of the remaining allele when a deletion occurs (35). In addition, genomic rearrangements can potentially act through effects of transvection (communication between alleles on homologous chromosomes), by deleting regulatory elements required for communication between alleles or simply by disrupting a coding sequence (36).

HIGH-THROUGHPUT METHODS FOR ANALYSIS OF GENOMIC DISORDERS

For many years the detection of submicroscopic (<5 Mb) rearrangements associated with genomic disorders required locus-specific testing, including fluorescence in situ hybridization (FISH), pulsed-field gel electrophoresis (PFGE), or multiplex ligation-dependent probe amplification (MLPA) techniques to resolve small genomic changes (<5 Mb) not detected by conventional microscopy. Rapid advancements in the development of microarray technologies have enabled high-resolution screening of the entire human genome.

The replacement of metaphase chromosomes as targets in conventional CGH with BAC or PAC (bacterial or P1 artificial chromosome) clones or oligonucleotide probes immobilized as arrays on glass slides enables genome-wide assessment of copy-number changes with a level of resolution depending only on the distance between the arrayed interrogating probes (37). Recently, array CGH proved also to be a powerful and promising method supplementing current routine diagnostic procedures in clinical cytogenetics (38).

In contrast to array CGH, SNP arrays have oligonucleotide specific to known SNPs. They do not require reference DNA from a healthy person. They offer the possibility to determine the parent of origin for a de novo deletion or duplication, establish paternity, and identify absence or loss of heterozygosity due to uniparental disomy (39).

CLINICAL CONSEQUENCES OF COPY-NUMBER VARIATIONS

In addition to visible CNVs due to chromosomal aberrations, such as trisomy 21 in patients with Down syndrome or monosomy X in women with Turner syndrome, many submicroscopic genomic deletion and duplication syndromes have been well known since the 1980s (40). Microdeletion of chromosome 15q11.2q12 as a cause of Prader-Willi syndrome was identified at Baylor College of Medicine in 1981 (41). Variable deletion breakpoints have been recognized in the majority of patients with α-thalassemia (42) and Duchenne muscular dystrophy (43), and recombination between red and green opsin genes, resulting in their abnormal copy number, was shown in 1986 (44). The recurrent common deletions flanked by LCRs in patients with steroid sulfatase deficiency were described a few years later (45, 46), and the first recurrent microduplication was identified in 1991 in patients with Charcot-Marie-Tooth disease type 1A (CMT1A) (47).

Recurrent Genomic Rearrangements

The gene- and LCR-rich genomic region of chromosome 17p11.2–p12 has been shown to be rearranged in a variety of different constitutional, evolutionary, and cancer-associated structural chromosome aberrations, and thus investigations of this region represent an excellent model to examine the potential role of genome architecture in DNA rearrangements (**Figure 1**) (16).

CMT1A/HNPP. Charcot-Marie-Tooth disease type 1A (CMT1A) and hereditary neuropathy with liability to pressure palsies (HNPP) are well-known genomic disorders caused, in the vast majority of cases, by copy-number change of the dosage-sensitive gene *PMP22*

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as a result of duplication (causing CMT1A) or reciprocal deletion (causing HNPP) of an ~1.4-Mb genomic fragment within 17p12. Both CMT1A and HNPP are peripheral neuropathies but with distinct clinical presentations. Whereas CMT1A is a chronic, insidious, slowly progressive neuromuscular disease with distal muscle wasting and weakness, HNPP presents with focal or multifocal neuropathy or can even manifest as a dominant carpal tunnel syndrome. Interestingly, ~90% of patients with sporadic Charcot-Marie-Tooth neuropathy had the common CMT1A duplication that arises as a de novo change (48), indicating a high frequency of genomic rearrangements in this region (4) (**Table 1**).

Smith-Magenis syndrome (SMS)/Potocki-Lupski syndrome (PTLS). SMS is a neurobehavioral disorder in which mental retardation (MR), sleep disturbance, stereotypical behaviors, and characteristic craniofacial features with multiple congenital anomalies result from haploinsufficiency of the *RAI1* gene on

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Table 1 Copy-number variations and neurogenetic disorders (expanded from References 49 and 50)

Syndrome	OMIM	Locus	Rearrangement	Gene(s)	Reference
Neurodevelopmental					
WBS del(7)q11.23	194050	7q11.23	del	CGS incl. *ELN*	51
dup(7)q11.23	609757		dup		52
AS	105830	15q11–q12	mat del, pat UPD15	*UBE3A*	53
PWS	176270		pat del, mat UPD15	CGS	54
dup(15)	608636	15q11–q13	dup	CGS	55
idic(15)		idic(15)(q13)	trip		56
MDLS	247200	17p13.3	del	CGS incl. *LIS1*	57
SMS	182290	17p11.2	del	CGS incl. *RAI1*	58
PTLS	610883		dup	*RAI1*	59
NF1	162200	17q11.2	del	CGS incl. *NF1*	60
del(17)q21.31	610443	17q21.31	del		61–63
DGS/VCFS	188400 192430	22q11.2	del	CGS incl. *TBX1*, *COMT*	64
dup(22)q11.2	608363		dup	CGS	65
del(22)q13	606232	22q13.3	del	*SHANK3/PROSAP2*	66
RTT	312750	Xq28	del	*MECP2*	67
Rett-like syndrome	300260		dup, trip	*MECP2*	68
PMD	312080	Xq22.2	dup, del	*PLP1*	69
Neurodegenerative					
PD	168601	4q21	dup, trip	*SNCA*	70
SMA	253300	5q13	del, gene conv	*SMN1*, *SMN2*	71
ADLD	169500	5q23.2	dup	*LMNB1*	72
CMT1A	118220	17p12	dup	*PMP22*	73, 74
HNPP	162500		del		
AD	104300	21q21	dup	*APP*	75

Abbreviations: AD, Alzheimer disease; ADLD, autosomal dominant leukodystrophy; AS, Angelman syndrome; CGS, contiguous gene deletion/duplication syndrome; CMT1A, Charcot-Marie-Tooth type 1 disease; del, deletion; dup(7)q11.23, reciprocal duplication of the WBS region; dup, duplication; gene conv, gene conversion; HNPP, hereditary neuropathy with liability to pressure palsies; MDLS, Miller-Dieker syndrome; NF1, neurofibromatosis type 1; PD, Parkinson disease; PMD, Pelizaeus-Merzbacher syndrome; PWS, Prader-Willi syndrome; RTT, Rett syndrome; SMA, spinal muscular atrophy; trip, triplication; UPD, uniparental disomy; WBS, Williams-Beuren syndrome.

chromosome 17p11.2 (76). In 70%–80% of SMS patients, *RAI1* is deleted along with the same ~3.7-Mb genomic segment flanked by large, complex LCRs (77).

The reciprocal 3.7-Mb duplication of this region has been described in patients with Potocki-Lupski syndrome (PTLS). Its clinical features are distinct from those seen with SMS, and include infantile hypotonia, failure to thrive, MR, syndromic autistic features, sleep apnea, and structural cardiovascular anomalies (**Table 1**).

Other common reciprocal microdeletion and microduplication syndromes. Several other recurrent microdeletions and reciprocal microduplications have been described. One example is chromosome 22q11.2 microduplication, reciprocal to deletion found in patients with DiGeorge and velocardiofacial syndromes; another is chromosome 15q11.2q12 microduplication, reciprocal to microdeletion found in patients with Prader-Willi and Angelman syndromes. In contrast to syndromic autism in patients with PTLS, patients with duplication 7q11.23, reciprocal to microdeletion found in patients with Williams-Beuren syndrome, present with speech delay and nonsyndromic autism spectrum behaviors (**Table 1**).

Recurrent rearrangement syndromes predicted and identified based on genome structure. Based on the knowledge of higher-order human genome architecture and the "rules" for the mechanism of NAHR (17), Sharp et al. (78) designed and developed a microarray with >2000 BAC clones specific for 130 genomic regions that were computationally predicted to be flanked by directly oriented LCRs. Using this approach, five novel recurrent pathogenic imbalances on chromosomes 1q21.1, 15q13, 15q24, 17q12, and 17q21.31 have been identified (63). All five microdeletions, and subsequently reciprocal microduplications, have been described in a large number of patients with different phenotypes, including autism, epilepsy, MODY (maturity-onset diabetes of the young), and schizophrenia, demonstrating that the paradigm shift of "genome first" is an efficient way of identifying new syndromes and the genetic bases of complex traits.

17q21.31 microdeletion syndrome. The chromosome 17q21.31 microdeletion syndrome is characterized by developmental delay (DD), hypotonia, facial dysmorphisms (including a long face, a tubular or pear-shaped nose, and a bulbous nasal tip), abnormally friendly behavior, epilepsy, heart defects, and urinary anomalies. It has been estimated to be responsible for up to 1% of MR cases. Interestingly, the 500–650-kb deletion flanked by LCRs arises only on a specific inversion haplotype H2, present in ~20% of the European population. Two candidate genes have been proposed as pathogenic, *CRHR1* (the corticotropin-releasing hormone receptor 1 gene) and *MAPT* (the microtubule-associated protein tau gene). A point mutation of *MAPT* is known to be responsible for frontotemporal dementia; however, no point mutation has been identified in individuals with MR (**Table 1**).

Nonrecurrent Genomic Rearrangements

Some nonrecurrent rearrangements have been detected as frequently as common recurrent microdeletions flanked and mediated by LCR/NAHR, suggesting some predisposing role of genomic architectural features.

Different sized (~0.2–2.6 Mb) microduplication CNVs involving the *MECP2* gene encoding X-linked methyl-CpG-binding protein 2 in Xq28 appear to be the most common nonrecurrent pathogenic subtelomeric microduplication (79). Deletion CNVs or loss-of-function mutations of *MECP2* are associated with Rett syndrome, a neurodevelopmental disorder that affects 1:10,000 girls. Males with *MECP2* duplications manifest DD/MR (100%), infantile hypotonia (100%), absent speech (84%), and a history of recurrent infections (80%). Genital or digital abnormalities, seizures, or lack of ambulation, sometimes followed by other

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ASD: autism spectrum disorder

sporadic signs, may be present in 40%–60% of the patients with *MECP2* duplication (80). In most cases, these microduplications are inherited from the mothers. Recently, Ramocki et al. (81) showed that the female carriers of *MECP2* duplications can manifest neurobehavioral and psychiatric symptoms. Mutations or deletions of *MECP2* in males are usually lethal. Using Haldane's rule (for a disease with a stable prevalence, the rate of appearance of mutated alleles must equal removal), and knowing the 1/3500 prevalence of Duchenne muscular dystrophy in male newborns, van Ommen (82) estimated the pathology-associated mutation rate of CNVs in the *DMD* gene as 1×10^{-4}.

COMPLEX TRAITS

Common complex traits, including autism, schizophrenia, epilepsy, Parkinson disease, Alzheimer disease, and emphysema, have been shown recently to result from CNVs in some fraction of patients. In Parkinson, Alzheimer, and epilepsy, the causative genes were initially identified through mapping and point-mutation analyses. However, for autism and schizophrenia, as for CMT1A, CNV first identified the locus of interest (**Table 1**).

Autism

Autism spectrum disorders (ASDs) (MIM 209850) comprise a heterogeneous group of neurodevelopmental abnormalities characterized by impairment of social interactions, problems in communication, and a restricted range of behaviors and interests. ASDs encompass five clinically defined conditions: autistic disorder, Asperger syndrome, disintegrative disorder, pervasive developmental disorder not otherwise specified (PDD-NOS), and Rett syndrome (DSM IV) (83, 84).

The genetic cause of ASDs is recognized only in ~10%–20% of cases. The most common cytogenetic abnormalities are maternally derived duplication of chromosome 15q11q13, found in 1%–3% of cases (84, 85); deletions of 7q22q31, 22q13.3, 2q37, 18q21q23, and Xp22; duplications of 7q11.23 and 17p11.2; and aneuploidy of chromosome Y. In addition, mutations in the *FMR1*, *MECP2*, *TSC1*, *NF1*, and *PTEN* genes that result in monogenic disorders—fragile X syndrome, Rett syndrome, tuberous sclerosis, neurofibromatosis, and PTEN hamartoma tumor syndrome, respectively—have been reported in patients with a syndromic ASD (85). Other genes, e.g., *CNTNAP2*, *SHANK3*, *NLGN3*, and *NLGN4*, that play an important role in neuronal activity, have been shown also to contribute to the etiology of ASDs (84, 85).

Several studies have shown the importance of CNVs in the etiology of ASDs. Sebat et al. (86) found de novo CNVs in 10% of simplex families with autism and in 2% of patients with an affected first-degree relative. Similar results have been obtained by Marshall et al. (84), who identified de novo CNVs in ~7% of patients with a sporadic ASD and in ~2% of families with affected siblings. Christian et al. (87) identified CNVs in 11.6% of patients with an ASD (14% de novo and 86% inherited); Morrow et al. (88) found submicroscopic de novo CNVs in >10% of patients with autism, and Weiss et al. (89) and Kumar at al. (90) reported common recurrent ~600-kb deletions and duplications of 16p11.2 in ~1% of patients with autism. In addition, genomic rearrangements involving *NRXN1* have been reported (91).

Using SNP microarrays, Glessner et al. (92) studied 859 + 1336 patients with ASD (and 1409 + 1110 healthy children in initial and replication cohorts, respectively) and identified that several new susceptibility genes encoding neuronal cell-adhesion molecules, including *NLGN1* and *ASTN2*, were enriched within CNVs in ASD cases compared to controls. They also found that CNVs within or surrounding genes involved in the ubiquitin pathways, including *UBE3A*, *PARK2*, *RFWD2*, and *FBXO40*, were affected by CNVs in ASD patients but not in controls.

Microdeletion 15q13.3. Two genomic regions flanked by LCRs, 1q21.1 and 15q13.3, have been shown recently to be responsible for

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complex neurobehavioral phenotypes, including ASDs. The 1.5-Mb critical region of the 15q13.3 deletion maps between BP4 and BP5 LCRs, distal to the Prader-Willi and Angelman syndrome region, and harbors six genes, including *CHRNA7* (92a). *CHRNA7* encodes the α7 subunit of the neuronal nicotinic receptor, which is a homopentameric synaptic ion channel protein.

Sharp et al. (93) reported nine individuals with a recurrent 15q13.3 microdeletion with MR, epilepsy, and variable facial and digital dysmorphisms. Helbig et al. (94) noted this deletion in 1% of patients with idiopathic epilepsy. Ben-Shachar et al. (95) described 20 patients with 15q13.3 deletion. They reported DD/MR or borderline IQ in 14 patients, of whom 6 also had an ASD. Other characteristics noted in this group of 20 were speech delay, aggressiveness, attention deficit/hyperactivity disorder (ADHD), epilepsy, other behavioral problems, and increased adoption and foster care. Intriguingly, six of the parents who apparently transmitted the deletion presented with MR, bipolar disorder, schizophrenia, and other antisocial behavior. Miller et al. (96) noted ADHD, anxiety disorder, and mood disorder in five patients with the BP4-BP5 deletion with minor dysmorphic features, significant expressive-language deficits, and a spectrum of neuropsychiatric impairments that include ASDs, anxiety disorder, and mood disorder. Van Bon et al. (97) studied 18 patients with deletion 15q13.3 and also observed a highly variable intra- and interfamilial phenotype with behavioral problems (including learning problems) in 10 of 17 patients. The phenotypic consequences of 15q13.3 duplication are currently uncertain.

Microdeletion and microduplication 1q21.1. Brunetti-Pierri et al. (98) reported 21 patients with a recurrent 1.35-Mb microdeletion and 15 patients with reciprocal microduplication that correlated with dysmorphic features, congenital anomalies, DD, and a spectrum of neuropsychiatric abnormalities, including learning disabilities, ADHD, autism, anxiety/depression, antisocial behavior, and aggression (even hallucinations). The deletion and duplication were associated with microcephaly and macrocephaly, respectively. Mefford et al. (99) identified similar deletion rearrangements in 25 patients with mild-to-moderate MR, microcephaly, cardiac abnormalities, and cataracts and duplication with MR or ASDs and other variable features. In addition, these phenotypes were subject to incomplete penetrance and variable expressivity. In the majority of cases, the 1q21.1 rearrangements were inherited. A careful clinical correlation of many families with similar genomic imbalance showed that the parents who carry the rearrangements, though initially considered to be healthy, upon extensive clinical evaluation present with an abnormal phenotype. A paralog of the *HYDIN* gene located on 16q22.2 (and inserted onto 1q21.1 during human evolution), which is implicated in autosomal recessive hydrocephalus, was proposed as a candidate gene for head-size abnormalities (98). This genomic region contains also the *DUF1220* sequences important to human brain and cognition function and behavior (100).

Schizophrenia

Interestingly, both 15q13.3 deletions and 1q21.1 deletions and duplications have been found in patients with schizophrenia. Schizophrenia is a severe psychiatric disorder characterized by hallucinations, delusions, cognitive deficits, and apathy, with a lifetime prevalence of ∼1%. Epidemiologic studies on twins indicate that schizophrenia has a complex genetic background with heritability estimated at 73%–90%.

Using high-density microarrays, two independent groups identified the 1q21.1 and 15q13.3 regions as schizophrenia loci. The International Schizophrenia Consortium (101) studied 3391 patients with schizophrenia (and 3181 controls); the other group, Stefansson et al. (102), investigated 1433 + 3285 schizophrenia patients (and 33,250 + 7951

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controls). Both groups identified well-known susceptibility CNVs: deletion 22q11.2, associated with DiGeorge syndrome and velocardiofacial syndrome (with neuropsychiatric manifestations in up to 30% of patients; 103, 104), and duplication 15q11.2.

Kirov et al. (105) and Rujescu et al. (106) have shown that altered function of *NRXN1* increases the risk of a number of neurodevelopmental disorders, including schizophrenia. Vrijenhoek et al. (107) screened 54 + 752 patients with schizophrenia and, in addition to *NRXN1*, they found three other genes affected by rare CNVs: *MYT1L*, *CTNND2*, and *ASTN2*.

Parkinson Disease

Parkinson disease (PD) is a neurodegenerative disease of the brain that often impairs motor skills and speech. It is a common disease, affecting up to 1% of the population over age 50. Several gene loci have been implicated in autosomal dominant, recessive, and X-linked forms of PD, and causative mutations have been found in seven genes: *SNCA*, *UCHL1*, *LRRK2*, *HTRA2* (autosomal dominant), *Parkin*, *DJ1* (autosomal recessive), and *SNCAIP* (sporadic). In addition, polymorphisms or mutations in the *HTRA2*, *LRRK2*, *NR4A2*, *NDUFV2*, *ADH3*, *FGF20*, *GBA*, and *MAPT* genes have been associated with susceptibility to PD.

The *SNCA* (*alpha-synuclein*) gene on chromosome 4q21 encodes a protein that is a major component of Lewy bodies, the intracellular pathological hallmark of typical PD and dementia with Lewy bodies. In addition to point mutations, different-sized nonrecurrent genomic duplications and triplications of the *SNCA* gene locus have been described in patients with PD. Interestingly, the triplications of *SNCA* are associated with a more severe, early-onset form of PD (108, 109), whereas the phenotype resulting from *SNCA* duplication resembles the typical late-onset idiopathic PD (110, 111). After screening 264 families with autosomal dominant PD, Ibáñez et al. (112) concluded that *SNCA* rearrangements may be even more frequent than point mutations in autosomal dominant PD. Supporting this notion, DNA variability within the *SNCA* promoter is associated with an increased risk of PD (113).

Alzheimer Disease

Alzheimer disease (AD) is the most common form of dementia. Typically, AD is diagnosed in people >65 years of age, although less-prevalent early-onset forms of AD have been reported. Histopathologically, AD is characterized by intracellular neurofibrillary tangles and extracellular amyloid plaques that accumulate in vulnerable brain regions. To date, 15 genomic loci have been linked to AD, and point mutations have been identified in three genes, *amyloid beta 4 precursor protein* (*APP*), *presenilin-1* (*PSEN1*), and *presenilin-2* (*PSEN2*) in patients with AD (OMIM 104310), early-onset AD3 (OMIM 607822), and familial AD4 (OMIM 606889), respectively.

Familial early-onset AD with cerebral amyloid angiopathy is caused by heterozygous mutations or different-sized nonrecurrent genomic duplications (580 kb to 6.37 Mb) of the dosage-sensitive *APP* gene, which encodes the amyloid precursor protein. Rovelet-Lecrux et al. (114) identified five different-sized duplication CNVs encompassing *APP* in 8% of patients that were point-mutation-negative for *APP*, *PSEN1*, and *PSEN2*. Duplication CNVs were not identified in 100 healthy controls with normal cognition over the age of 60 years. Interestingly, not only missense mutations of *APP* but also genetic variations in *APP* promoter sequences are associated with AD (115). Perhaps even more intriguing, early-onset AD is part of the phenotype of Down syndrome due to trisomy 21, the chromosome on which *APP* is located (21q21.3).

Epilepsy

It has been estimated that up to 40% of epilepsies are genetically determined and inherited in

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a Mendelian, non-Mendelian, or complex-trait fashion. In rare monogenic epilepsies, most of the genes are subunits of neuronal voltage- or ligand-gated ion channels and encode proteins related to neuronal maturation and migration during embryonic development.

Inactivating point mutations and genomic deletions on chromosome 2q24.3 leading to haploinsufficiency of the *SCN1A* gene encoding an alpha subunit of the sodium channel have been found in patients with GEFS+ (generalized epilepsy with febrile seizures plus) and SMEI (severe myoclonic epilepsy of infancy) or Dravet syndrome (116). Helbig et al. (94) identified the 15q13.3 microdeletions encompassing the *CHRNA7* gene in 12 of 1223 individuals with idiopathic generalized epilepsy (not detected in 3699 controls) and suggested that 15q13.3 microdeletions constitute the most prevalent risk factor for common epilepsies identified to date. Using array CGH, Erez et al. (117) identified variable-sized deletion CNVs involving exons 1–4 of the *CDKL5* gene in three females with early-onset seizures. Point mutations in *CDKL5* have been reported in 10% of females with a Rett-sydrome-like phenotype and early-onset seizures.

Pancreatitis

Mutations in the cationic trypsinogen gene *PRSS1* (encoding cationic trypsinogen) and the *SPINK1* gene have been found in patients with chronic pancreatitis. In five unrelated families with hereditary idiopathic pancreatitis, a similar-sized 605-kb duplication and triplication containing the entire *PRSS1* gene and the *PRSS2* gene (encoding anionic trypsinogen) have been identified (**Table 2**).

Developmental Delay/Mental Retardation

The overall rate of detection of genomic rearrangements in children with DD/MR and/or multiple congenital anomalies is 12%–18%, with 3%–5% detected by banded karyotype, 5%–6% detected by subtelomeric FISH but not karyotype, and 4%–7% detected by array CGH (higher for tiling-path than for targeted arrays) but not by karyotype or subtelomeric FISH. This would equate to at least 9%–13% MR-associated CNVs detected by array CGH but not karyotype, because array CGH is superior to subtelomeric FISH for detecting telomeric abnormalities (**Figure 2**) (38).

Table 2 Structural variations and other traits

Trait	Gene/locus	Reference
Psoriasis	*beta-defensin*	118
Emphysema	5p15.33, 7q22.1	119
Acute myocardial infarction and stroke (smokers)	*C4B* encoding C4, the fourth component of complement copy	120
Alveolar capillary dysplasia	*FOXF1*	121
Hypertrichosis	17q24.2-q24.3	122
Chronic pancreatitis	*PRSS1*	123
Lupus with glomerulonephritis	*FCGR3B*	124
Crohn disease	*HBD-2 (beta-defensin)*, *IRGM (immunity-related GTPase family, M)*	125, 126
Heterozygous familial hypercholesterolemia	*LDLR*	127
Atherosclerosis	*LPA*	127
Hemophilia A	*Factor VIII*	128
Malaria susceptibility	*a-globin*	129
Systemic lupus erythematosus	*Complement C4 component*, *FCGR3B*	130, 131
HIV susceptibility	*CCL3L1*	132

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Figure 2

An ideogram of human chromosomes with selected genomic loci responsible for syndromic or nonsyndromic mental retardation. Duplication CNVs are shown in red and deletion CNVs are in blue. For further details, see **http://www.bcm.edu/geneticlabs/cma/index.html**.

Susceptibility to Other Diseases

CNVs also have been associated with susceptibility to HIV infection, Crohn disease, glomerulonephritis, psoriasis, systemic lupus erythematosus, emphysema, and many other human diseases (**Table 2**).

SUMMARY

CNVs have a higher per-locus mutation rate than SNPs and are likely responsible for a significant amount of both human gene and genome evolution, via gene duplication, triplication, and exon shuffling, as well as genetic diversity between individuals—perhaps to a larger extent than SNPs. Sporadic Mendelian traits [including birth defects (133)], common (complex) traits (including autism, schizophrenia, and MR), and disease susceptibility can result from CNVs. Both recombination-based (NAHR and NHEJ) and replication-based (FoSTeS/MMBIR) mechanisms lead to formation of CNVs.

FUTURE ISSUES

1. Microarray technology applied in the clinic will identify CNVs as the basis of many human traits. This will enable functional mapping of the human genome, including potentially assigning function to many genes. The forward genomic screen of human clinical phenotypes will identify clinically relevant CNVs/structural variations, and a reverse-genomics analysis of patients harboring such rearrangements will facilitate the characterization of the phenotypic consequences of such mutational changes.
2. CNVs may be responsible for many different behavioral traits.
3. To what extent will CNV correlate with transcriptional changes of the individual genes whose copy number is changed?
4. Will CNVs provide novel routes to therapy?

DISCLOSURE STATEMENT

J.R.L. is a consultant for Athena Diagnostics, 23andMe, and Ion Torrent Systems. The authors are based in the Department of Molecular and Human Genetics at Baylor College of Medicine, which offers extensive genetic laboratory testing, including use of arrays for genomic copy number analysis, and derives revenue from this activity.

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